Md Mustakin Alam

Ph.D. Student in Computer Science
University of Louisiana at Lafayette
School of Computing and Informatics

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RESEARCH INTERESTS

My research interests are at the intersection of Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision, with a focus on efficient, adaptive, and trustworthy Deep Learning architectures.

EDUCATION

Ph.D. in Computer Science Aug 2024 – May 2028
University of Louisiana at Lafayette
GPA: 4.00 / 4.00
Supervisor: Dr. Aminul Islam

Master of Science in Computer Science Aug 2024 – May 2026
University of Louisiana at Lafayette
GPA: 4.00 / 4.00

Bachelor of Science in Computer Science and Engineering May 2019 — May 2023
BRAC University
CGPA: 3.96 / 4.00

SELECTED PUBLICATIONS

1. **Md Mustakin Alam**, Shaker Islam, Aminul Islam. **"Batch Pruning by Activation Stability"**. In: The Fourteenth International Conference on Learning Representations, ***ICLR 2026***. paper | code

2. Aminul Islam, **Md Mustakin Alam**, Shaker Islam. **"The Birth of Vision Language"**. In: 33rd ACM International Conference on Multimedia, ***ACMMM 2025***. paper | code

3. **Md Mustakin Alam**, Tanjim Ahmed, Meraz Hossain, Mehedi Hasan Emo, Md. Kausar Islam Bidhan, Md Tanzim Reza, Md. Golam Rabiul Alam, Mohammad Mehedi Hassan, Francesco Pupo, Giancarlo Fortino. **"Federated Ensemble-Learning for Transport Mode Detection in Vehicular Edge Network"**. In: Future Generation Computer Systems, ***FGCS (2023)***. paper | code

4. Md Mahadi Hasan, David Dew Mallick, Towhid Khan, **Md Mustakin Alam**, Md. Humaion Kabir Mehedi, Annajiat Alim Rasel. **"SweetCoat-2D: Two-Dimensional Bangla Spelling Correction and Suggestion Using Levenshtein Edit Distance and String Matching Algorithm"**. In: 2023 International Joint Conference on Neural Networks, ***IJCNN 2023***. paper

TEACHING EXPERIENCE

Graduate Teaching Assistant August 2024 – Present
University of Louisiana at Lafayette

- Course: CMPS340 Design and Analysis of Algorithms by Dr. Aminul Islam
- Provided consultations during office hours
- Graded homeworks and exams

Adjunct Lecturer September 2023 – July 2024

BRAC University

- Taught Courses:
  CSE422 Artificial Intelligence (Theory)
  CSE422 Artificial Intelligence (Laboratory)
  CSE341 Microprocessors (Laboratory)
  CSE461 Introduction to Robotics (Laboratory)
- Supervised student research and projects

Undergraduate Teaching Assistant February 2022 – April 2023

BRAC University

- Assigned Courses:
  CSE110 Programming Language I
  CSE111 Programming Language II
  CSE424 Pattern Recognition
  CSE431 Natural Language Processing
  CSE449 Parallel, Distributed, and High-Performance Computing
- Graded homeworks
- Provided consultation hours for students and assisted them with problem-solving and exam preparation
- Introduced students to research methodology and guided them in writing research papers

PROFESSIONAL EXPERIENCE

Machine Learning Engineer June 2023 – July 2024

mPower Social Enterprises Ltd.

- Developed machine learning models for real-world applications
- Built LLM-based chatbot and question generation systems
- Developed pregnancy risk prediction model using healthcare data
- Worked on speech-to-text systems by fine-tuning Whisper models

HONORS AND AWARDS

- Graduate Student Supplements (GRASS) Award – NSF & Louisiana Board of Regents 2026
- Academic Excellence Award – University of Louisiana at Lafayette 2025 – 2026
- Graduate Student Research Showcase Poster Competition – 2nd place 2025
- Vice Chancellor's List – BRAC University (Multiple semesters) 2019 – 2023
- Dean's List – BRAC University 2021
- Bachelor's with Highest Distinction – BRAC University 2023

EXPERIENCE WITH RESEARCH GRANTS

- Vision Language, NSF Future CoRe. *Under Review* September 2025
- Energy-Efficient Deep Learning Training via Dynamic Pruning,
  NSF Future CoRe. *Under Review* February 2026

ADVISING AND MENTORING

- Shaker Islam (Bachelor of Science, BRAC University)
  - Co-authored articles at ACM MM 2025 and ICLR 2026.
- Sébastien Ricart and Lou-Anne Lecocq (Visiting Intern from CESI, UL Lafayette)
  - Supervised visiting intern projects in 2025.
- Louis Covemacker and Hugo Kergoat (Visiting Intern from CESI, UL Lafayette)
  - Supervised visiting intern projects in 2024.
- A.S.M Mehedi Hasan, Md. Alvee Ehsan, Kefaya Benta Shahnoor, and Syeda Sumaiya Tasneem (Bachelor of Science, BRAC University)
  - Co-supervised undergraduate thesis in 2023.

TECHNICAL SKILLS

Programming: Python, Java, C, PHP, Bash, SQL
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
Tools: Linux, Git, Jupyter, Google Colab, Overleaf
Other: Arduino, Raspberry Pi

ACADEMIC AND EXTRACURRICULAR SERVICES

- Reviewer, 59^{th} Hawaii International Conference on System Sciences (HICSS 2026)
- Reviewer, Evolving Systems (Springer)
- Science Fair Judge – University of Louisiana at Lafayette (2025, 2026)
- Science Day Volunteer – University of Louisiana at Lafayette (2025)
- Student Mentor – BRAC University (2021 – 2022)
- Senior Executive – Robotics Club, BRAC University
- Assistant Director – Entrepreneurship Development Forum, BRAC University

PRESENTATIONS AND TALKS

Conference Talk	Batch Pruning by Activation Stability, *The Fourteenth International Conference on Learning Representations (ICLR)*, Recorded Online, March 2026
Poster	Energy-Efficient Deep Learning Training via Dynamic Data Pruning, *Graduate Student Research Showcase Poster Competition*, University of Louisiana at Lafayette, March 2026
Poster	Vision Language, *Graduate Student Research Showcase Poster Competition*, University of Louisiana at Lafayette, March 2025
Invited Talk	Vision Language, *Graduate Seminar*, University of Louisiana at Lafayette, March 2025
Conference Talk	Bidirectional LSTM and NLP-Based Sentiment Analysis of Tweets, *14th Soft Computing and Pattern Recognition (SoCPaR'22)*, Online Presentation, December 2022